Fijnman-GSA^3: A Fail-Closed Framework for Structural Reasoning

(v5.1.1: Layered Symbolic Core with Constitutional Grounding)

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Abstract

Contemporary reasoning systems exhibit persistent instability: identical inputs can yield multiple, mutually incompatible outputs depending on interpretation, hidden assumptions, or implicit completion of missing information. This instability is not incidental but structural: it arises from permitting inference under under-specification.

We present a fail-closed framework for reasoning in which admissibility replaces truth as the primary criterion for existence. A statement may exist only if it is fully specified and cannot be structurally invalidated by alternative interpretations. Any attempt to infer beyond explicitly defined variables and relations results in a mandatory halt.

In this version (v5.1.1), we extend the framework with a layered symbolic core grounded in a constitutional quotient. State spaces are represented symbolically via constraints, and invariants are organised into a hierarchy $\Phi_0 \subseteq \Phi_1 \subseteq \cdots \subseteq \Phi_k$. Knowledge is defined level-dependently as the maximal Φ_i-consistent residue of a successor set. This structure avoids enumerative explosions, handles uncertainty without probability, and preserves fail-closed semantics. A mode profile replaces the single mode classification, and admissibility is determined by the highest non-empty kernel level I^*. The constitutional quotient $\hat{S} = S/{\equiv_{\Phi_0}}$ provides a clean topological foundation.

All extensions preserve the fail-closed nature of the framework. The categorical analysis of reachability (v4.9) remains as a complementary layer, motivating enriched morphisms without altering the core validation pipeline.

1 Introduction

Reasoning systems—both human and computational—routinely produce divergent outputs from identical inputs. Given a dataset D and a method M, it is common that

$$M_1(D) \to O_1, \qquad M_2(D) \to O_2, \quad \text{with } O_1 \neq O_2.$$

Such divergence is typically attributed to differences in models, perspectives, or interpretation. However, this explanation is incomplete. The deeper issue is that most systems permit inference under under-specified conditions.

When essential variables, constraints, or relations are not explicitly defined, systems compensate by introducing implicit assumptions. These assumptions may remain unexamined, yet they directly influence outcomes. As a result, reasoning becomes dependent on hidden completions rather than explicit structure.

This paper adopts a different approach. Instead of attempting to improve inference under under-specification, we prohibit it.

We construct a fail-closed framework in which reasoning is only permitted when all required components are explicitly specified. Missing definitions, ambiguous mappings, or competing interpretations are not resolved through approximation or estimation, but trigger a mandatory halt condition.

Within this framework, existence is redefined: a statement does not exist because it is asserted or interpretable, but because it is admissible under a fully specified structure. Admissibility replaces truth as the primary criterion. A statement is admissible if and only if it can be evaluated without introducing new assumptions.

Reasoning is modeled as a sequence of admissible state transitions within a constrained system. Each transition must preserve a set of invariants; any violation results in rejection. Knowledge is not defined as belief or justified truth, but as the set of statements that remain invariant across all admissible continuations.

A central component of the framework is a convergence requirement: iterative refinement must approach a stable limit without the introduction of new assumptions. If refinement requires additional assumptions, the process diverges and is terminated. This establishes a structural boundary against inference under under-specification.

1.1 Contribution of Version 5.1.1

Version 5.1.1 replaces the enumerative state space and flat invariant set with a *layered symbolic core* and introduces a *constitutional quotient* that separates structural grounding from epistemic refinement:

- **Symbolic state space**: $S = \{\sigma \mid \Psi(\sigma)\}$ where Ψ is a logical formula; states are assignments to variables.
- **Constitutional quotient**: $\hat{S} = S/{\equiv_{\Phi_0}}$ where Φ_0 are minimal structural invariants. All topological and dynamic operations are defined on $\hat{S}$.
- **Invariant hierarchy**: $\Phi_0 \subseteq \Phi_1 \subseteq \cdots \subseteq \Phi_k$, with Φ_0 constitutional and Φ_i $(i > 0)$ epistemic refinements.

- **Level-relative kernel**: $K^{(i)}(S') = S'$ if S' is Φ_i-consistent, otherwise $\emptyset$. The kernel is the maximal Φ_i-consistent *residue* of S'. This definition deliberately avoids selecting maximal consistent subsets; any internal inconsistency collapses the entire set.
- **Admissibility**: The highest level $I^* = \max\{i \mid K^{(i)}(S') \neq \emptyset\}$ determines the outcome.
- **Mode profile**: A vector of modes per level, replacing the single mode classification.

These extensions preserve fail-closed semantics while enhancing scalability and handling uncertainty without probability. The categorical analysis of reachability (v4.9) remains as a complementary layer, motivating enriched morphisms without altering the core validation pipeline.

2 Constitutional Definitions

The framework is built on a layered symbolic core. All definitions are mechanistically executable: every step reduces to formula manipulation or solver invocation, and all choices are deterministic.

2.1 Symbolic State Space

A state space is defined by a formula Ψ in a decidable logic (e.g., QF_LIA, QF_UF):

$$S = \{\sigma \mid \Psi(\sigma)\},$$

where σ ranges over assignments to variables. Ψ must be fully specified. If variables are missing, constraints are ambiguous, or the logic falls outside the specified class:

$$\mapsto \textsc{Halt_Spec_Required}.$$

2.2 Constitutional Quotient

All topological and dynamic operations are defined on the quotient space induced by the constitutional invariants Φ_0:

$$\hat{S} = S/\equiv_{\Phi_0},$$

where $\equiv_{\Phi_0}$ identifies states with identical Φ_0-values. Symbolic state definitions and successor constraints are interpreted in $\hat{S}$ after quotienting; the quotient map is denoted $\pi : S \to \hat{S}$.

2.3 Invariant Hierarchy

A finite, totally ordered family of invariants refines the constitutional base:

$$\Phi_0 \subseteq \Phi_1 \subseteq \cdots \subseteq \Phi_k.$$

Φ_0 are the minimal, always-required invariants (the constitutional layer). For $i > 0$, Φ_i are stricter epistemic refinement layers (finer distinctions).

2.4 Symbolic Substate

A successor set $S' \subseteq \hat{S}$ is represented symbolically as:

$$S' = \{\sigma \mid \Psi(\sigma) \wedge \Xi(\sigma)\},$$

where Ξ is an additional constraint in the same logic as Ψ. If $\Psi \wedge \Xi$ is unsatisfiable, then $S' = \emptyset$.

2.5 Φ_i-Equivalence and Consistency

For level i:

$$\sigma \sim_{\Phi_i} \sigma' \iff \forall \phi \in \Phi_i : \phi(\sigma) = \phi(\sigma').$$

The equivalence class of σ is the logical formula:

$$[\sigma]_{\Phi_i} = \{\tau \mid \Psi(\tau) \wedge \bigwedge_{\phi \in \Phi_i} \phi(\tau) = \phi(\sigma)\}.$$

S' is Φ_i-*consistent* if:

$$\forall \sigma_1, \sigma_2 \in S' : \forall \phi \in \Phi_i : \phi(\sigma_1) = \phi(\sigma_2).$$

Symbolically, this is equivalent to:

$$\forall \tau_1, \tau_2 : (\Psi(\tau_1) \wedge \Xi(\tau_1) \wedge \Psi(\tau_2) \wedge \Xi(\tau_2)) \Rightarrow \bigwedge_{\phi \in \Phi_i} (\phi(\tau_1) = \phi(\tau_2)).$$

If this formula is valid in the chosen logic, S' is Φ_i-consistent.

2.6 Level-Relative Kernel

$$K^{(i)}(S') = \begin{cases} S' & \text{if } S' \text{ is } \Phi_i\text{-consistent,} \\ \emptyset & \text{otherwise.} \end{cases}$$

Interpretation. The kernel is the maximal Φ_i-consistent *residue* of S': when S' is consistent it coincides with S' itself; otherwise it is empty, indicating that no common Φ_i-value exists across all $\sigma \in S'$. This definition deliberately avoids selecting maximal consistent subsets; any internal inconsistency collapses the entire set.

2.7 Kernel Monotonicity

Theorem 2.1 (Kernel Monotonicity)**.** *For all $i < j$:*

$$K^{(j)}(S') \subseteq K^{(i)}(S').$$

Proof. Because $\Phi_i \subseteq \Phi_j$, Φ_j-consistency implies Φ_i-consistency. Two cases:

- If S' is Φ_j-consistent, then it is also Φ_i-consistent, so $K^{(j)}(S') = S' = K^{(i)}(S')$.
- If S' is not Φ_j-consistent, then $K^{(j)}(S') = \emptyset \subseteq K^{(i)}(S')$.

Thus the inclusion holds in all cases. □

2.8 Admissibility Criterion (I^*)

$$I^* = \max\{\, i \mid K^{(i)}(S') \neq \emptyset \,\}.$$

If I^* exists:

$$\mapsto \textsc{Admissible_Level}(I^*).$$

If no level yields a non-empty kernel:

$$\mapsto \textsc{Refusal}.$$

The index set is finite and totally ordered, so I^* is well-defined (or absent).

2.9 Knowledge

$$\text{Knowledge}_i(S') = K^{(i)}(S').$$

Knowledge is level-dependent; the system reports the highest level with non-empty residue.

2.10 Modes per Level

For each level i, given $S' = \{\sigma \mid \Psi(\sigma) \wedge \Xi(\sigma)\}$:

- Strict: exactly one model of $\Psi \wedge \Xi$ exists, and uniqueness is decidable in the chosen logic. Uniqueness is checked via solver-based disequality test: $\exists! \sigma : \Psi(\sigma) \wedge \Xi(\sigma)$ is equivalent to $\forall \sigma_1, \sigma_2 : (\Psi(\sigma_1) \wedge \Xi(\sigma_1) \wedge \Psi(\sigma_2) \wedge \Xi(\sigma_2)) \Rightarrow \sigma_1 = \sigma_2$.
- Safe: at least two distinct models exist, and S' is Φ_i-consistent.
- Defeasible: S' is not Φ_i-consistent.

These cases are mutually exclusive and tested in the listed order. If uniqueness cannot be determined (solver returns Unknown or the theory does not support uniqueness checking):

$$\mapsto \textsc{Refusal}.$$

2.11 Mode Profile

$$\text{ModeProfile}(S') = (m_0, m_1, \ldots, m_k),$$

where each m_i is the mode of level i. The profile is descriptive only; admissibility is determined solely by I^*.

2.12 Dynamic Updates

An update u selects a surviving subset $\hat{S}_u \subseteq \hat{S}$ (the quotient space). The reachability order is restricted accordingly:

$$R_A^u = R_A \cap (\hat{S}_u \times \hat{S}_u), \qquad \leq_u = (R_A^u)^*.$$

If $\hat{S}_u = \emptyset$, the system halts with Refusal, preserving fail-closed semantics.

For any successor set $S' \subseteq \hat{S}_u$, the kernel after update is:

$$K^{(i),u}(S') = K^{(i)}(S') \cap \hat{S}_u,$$

where the intersection is taken after first restricting S' to $\hat{S}_u$ (i.e., S' is already understood as a subset of $\hat{S}_u$).

2.13 Decidability and Solver Behavior

Situation	Outcome
Logic outside specified class	Halt_Spec_Required
Solver returns Unknown	Refusal (treated as semantic n
Query well-formed but not decidable by the adopted procedure	Refusal

2.14 Axioms

The following axioms govern the transition structure.

Axiom 2.1 (Invariant Preservation)**.** *For every admissible transition $(\sigma, \tau) \in R_A$ and every invariant $\phi \in \Phi_0$ (the constitutional invariants), we have $\phi(\sigma) = \phi(\tau)$. Invariants at higher levels are preserved by definition of consistency.*

Axiom 2.2 (Separation)**.** *If $\sigma \leq \tau$ and $\tau \leq \sigma$ (where $\leq$ is the reflexive-transitive closure of R_A), then $[\sigma]_{\Phi_0} = [\tau]_{\Phi_0}$. In the quotient $\hat{S} = S/{\equiv_{\Phi_0}}$, this ensures antisymmetry of the reachability order.*

3 Relational Admissibility: RGR and RCR

The relational admissibility gates are applied sequentially as in v4.9.1, now interpreted in the symbolic setting and referring to the constitutional invariants Φ_0.

3.1 Relational Grounding Requirement (RGR)

A state σ is relationally grounded iff its relational component $R_\sigma \neq \emptyset$. Inference is admissible only over relationally grounded states. Formally, the state space formula must satisfy:

$$\Psi(\sigma) \Rightarrow R_\sigma \neq \emptyset.$$

At the proposal level, failure of relational grounding yields:

$$R = \emptyset \Rightarrow A_{\text{pre}}(q) = \textsc{Halt_Spec_Required}.$$

At the state level, any attempted representation with $R_\sigma = \emptyset$ fails completeness and therefore does not belong to S.

3.2 Relational Closure Requirement (RCR)

A relationally grounded state σ satisfies the Relational Closure Requirement iff its relational structure is consistent with the minimal invariant family Φ_0:

$$\Delta\Phi_0(\sigma) = 0.$$

A state that is grounded but fails closure is classified as inconsistent:

$$R_\sigma \neq \emptyset \wedge \Delta\Phi_0(\sigma) \neq 0 \Rightarrow A(q, \sigma) = \textsc{Reject}.$$

3.3 Admissibility Pipeline

For a proposal q and a state σ that results from mapping q to σ (when possible), the admissibility outcome is determined as follows:

- $A_{\text{pre}}(q) = \textsc{Halt_Spec_Required}$ no state is formed; system halts.
- Otherwise, $R_\sigma \neq \emptyset$, $\Delta\Phi_0(\sigma) \neq 0$ $A(q, \sigma) = \textsc{Reject}$.
- Otherwise, $R_\sigma \neq \emptyset$, $\Delta\Phi_0(\sigma) = 0$ $A(q, \sigma) = \textsc{Admissible}$ (proceed to transition).

Grounding permits evaluation. Closure permits admissibility. The kernel is what survives both.

4 Dynamics and Knowledge

4.1 Trajectory

$$\sigma_0 \to \sigma_1 \to \sigma_2 \to \dots$$

4.2 Properties

- **Linearity**: no inadmissible return to a prior state without new admissible information.
- **Convergence**: $\lim_{n\to\infty} \sigma_n = \sigma^*$ (if a unique fixed point exists).
- **Divergence**: if no stable kernel exists, $\to$ Refusal.

4.3 Knowledge Object

A set $K \subseteq S$ is forward-closed under admissible transitions if

$$\forall \sigma \in K, \forall q \text{ admissible} : \Theta(q, \sigma) \subseteq K.$$

4.4 Knowledge as Level-Relative Kernel

Knowledge is not a single set but a family indexed by invariant level:

$$\text{Knowledge}_i(S') = K^{(i)}(S').$$

Knowledge is level-dependent; there is no global truth, only structural invariant residue. The system reports the highest level I^* at which a non-empty kernel exists.

5 Modes and Mode Profile

5.1 Mode per Level

For each level i, given $S' = \{\sigma \mid \Psi(\sigma) \wedge \Xi(\sigma)\}$:

- Strict: there exists exactly one σ satisfying $\Psi \wedge \Xi$, and uniqueness is decidable in the chosen logic. Uniqueness is checked via solver-based disequality test: $\exists! \sigma : \Psi(\sigma) \wedge \Xi(\sigma)$ is equivalent to $\forall \sigma_1, \sigma_2 : (\Psi(\sigma_1) \wedge \Xi(\sigma_1) \wedge \Psi(\sigma_2) \wedge \Xi(\sigma_2)) \Rightarrow \sigma_1 = \sigma_2$.
- Safe: there exist at least two distinct models of $\Psi \wedge \Xi$, and S' is Φ_i-consistent.
- Defeasible: S' is not Φ_i-consistent.

These cases are mutually exclusive and tested in the order listed.

Decidability for Strict. If uniqueness cannot be determined (solver returns Unknown, or the theory does not support uniqueness checking):

$$\mapsto \text{Refusal}.$$

5.2 Mode Profile

$$\text{ModeProfile}(S') = (m_0, m_1, \ldots, m_k)$$

where each m_i is the mode of level i. The profile fully describes the system state.

Descriptive status. ModeProfile is descriptive only and does not alter admissibility, which is determined exclusively by $I^* = \max\{i \mid K^{(i)}(S') \neq \emptyset\}$.

6 Multi-Agent Extension

This section remains largely unchanged from v4.9.1, with the adaptation that each agent's state space is symbolic and invariants may be layered. The joint kernel is defined as the intersection of per-agent kernels over all levels.

6.1 Agent Definition

An agent $a \in \mathcal{A}$ has:

- Individual state space $S_a \subseteq S$ (symbolic)
- Proposal set $Q_a \subseteq Q$
- Admissibility operator $A_a : Q_a \times S_a \to \Omega$
- Guard $G_a : S_a \times Q_a \to \{0, 1\}$
- Individual invariants Φ_a (a hierarchy)

6.2 Joint State and Proposals

Joint state:

$$\Sigma = ((S_a)_{a \in \mathcal{A}}, C, \mathcal{H})$$

Joint proposal:

$$\mathbf{q} = (q_1, \ldots, q_n) \in Q_{\text{joint}} = \prod_{a \in \mathcal{A}} Q_a$$

Joint admissibility:

$$A_{\text{joint}}(\mathbf{q}, \Sigma) = \textsc{Admissible} \iff \forall a : A_a(q_a, S_a) = \textsc{Admissible}$$

Joint guard:

$$G_{\text{joint}}(\Sigma, \mathbf{q}) = 1 \iff \forall a : G_a(S_a, q_a) = 1$$

6.3 Multi-Agent Invariants

$$\Phi_{\text{joint}} = \bigcup_a \Phi_a \cup \{\phi_{\text{comm}}, \phi_{\text{cons}}, \phi_{\text{priv}}\}$$

- ϕ_{comm}: communication is explicit, traceable, and admissible
- ϕ_{cons}: shared facts are consistent across all agents
- ϕ_{priv}: private information is shared only with explicit consent

6.4 Joint Kernel

$$K_{\text{safe,joint}}(\mathbf{q}, \Sigma) = \bigcap_{a \in \mathcal{A}} K_{\text{safe},a}(S'_a)$$

The joint kernel is level-wise, and I^* is computed as the maximum level where all agents have non-empty kernels.

Full formalisation is left for future work.

7 Relation to Epistemic Logic

This section is unchanged from v4.9.1; it highlights the conceptual shift from possibility to admissibility.

7.1 Common Ground

Both epistemic logic and this framework are concerned with knowledge, information, and dynamics under updates or transitions.

7.2 Key Differences

Possibility vs Admissibility Classical epistemic logic asks: what is possible? Knowledge is defined as truth across all accessible possible worlds. This framework asks: what is admissible? Knowledge is invariance under admissible transitions.

Updates vs Validation Dynamic epistemic logic models knowledge updates as operations on epistemic models. This framework separates generation from admissibility: proposals are validated against explicit invariants.

Constraints vs Possible Worlds Epistemic logic works with possible worlds and accessibility relations. This framework works with explicit invariants Φ and admissible transitions Θ (equivalently, the induced relation R_A).

Common Knowledge as Fixed Point In dynamic epistemic logic, common knowledge is the fixed point of mutual knowledge. In this framework, common knowledge corresponds to the fixed point of the joint kernel:

$$K_{\text{safe,joint}} = \bigcap_a K_{\text{safe},a}.$$

7.3 Mapping (Not Identity)

- Kripke worlds $\to$ states S
- Accessibility relations $\to$ admissible transitions R_A
- Common knowledge $\to$ joint kernel fixed point

7.4 Why the Shift

The shift from possibility to admissibility is motivated by applications in which underspecification must be handled explicitly rather than silently completed.

8 Test Validation

The test validation section is unchanged from v4.9.1; it demonstrates the behaviour on classic puzzles, all of which can be encoded in the layered symbolic core. The kernel shrinkage in muddy children now corresponds to increasing the invariant level.

8.1 Problems Tested

Problem	**Type**	**Demonstration**
Muddy children	Knowledge acquisition	Kernel shrinks stepwise under SAFE/DEFI
Coordinated attack	Uncertainty, communication	No false certainty produced; uncertainty re
Cheating students	Privacy, implicit assumptions	ϕ_{priv} requires explicit consent
Fitch's paradox	Knowability, contradiction	Explicit knowability required; paradoxical c
Surprise examination	Paradox, self-reference	Kernel shrinks; if empty $\to$ REFUSAL; par

8.2 What the Tests Demonstrate

1. No output without specification: incompleteness $\to$ HALT_SPEC_REQUIRED; ambiguity $\to$ HALT_SPEC_REQUIRED.
2. No probabilistic assumptions: the kernel is the intersection of all admissible paths; no guessing, no fallback.

3. No construction of paradoxical states: explicit specification is required; kernel shrinkage stops at REFUSAL.

4. No inconsistency tolerated: consistency invariants enforce consistency; violation $\rightarrow$ REFUSAL.

8.3 What the Tests Do Not Prove

- No proof that the framework solves all conceivable problems.
- No proof that the framework produces only correct outcomes (this is not enforceable by definition).
- No proof that the specifications $(\Phi, \text{modes}, \text{invariants})$ are optimal.

What is demonstrated is consistency under explicit specifications and absence of hallucination within the tested scenarios.

9 Empirical Stability Under Composition

This section is unchanged; it reports observations on LLM composition under fail-closed constraints. The symbolic layered core strengthens these observations by providing a formal basis for bounded drift.

10 Scope and Limitations

The framework presented here is a structural validation system, not a general reasoning engine.

10.1 What the System Does

The system determines whether a proposal is admissible given:

- a fully specified symbolic state space Ψ
- a complete set of invariant hierarchies Φ_i
- a well-defined transition operator Θ (generating symbolic S')

It validates, filters, and executes admissible transitions. It does not generate proposals, infer missing specifications, or assign probabilities.

10.2 What the System Does Not Do

- Generate content: proposals must be supplied externally.
- Handle probability: uncertainty is modeled via kernel emptiness at higher levels, not distributions.
- Seek completeness: if specification is missing, the system halts.
- Optimize: there is no utility function and no preference ordering.
- Explain: the system does not provide causal or intentional explanations.

10.3 Boundaries

The system is restricted to validation, while generation, specification, and interpretation remain external. All content, including Ψ, Φ_i, and proposals, is supplied externally. The system only checks consistency, completeness, and convergence.

10.4 Applicability

The framework is suited for domains where:

- structural constraints dominate over intentions or narratives
- admissibility is more relevant than possibility
- underspecification must be handled by halting, not guessing

10.5 Limitations

The framework does not claim to:

- replace existing logics (DEL, HoTT, etc.)
- solve all reasoning problems
- guarantee correctness of external inputs
- provide a complete theory of knowledge

Its strength lies in structural discipline, not in expressive richness. It is a filter, not a generator.

11 Topological Grounding of Admissible Reasoning

The definitions of S (symbolic), R_A, Φ_i, and the kernel admit a natural topological interpretation. All states in S are relationally grounded ($R_\sigma \neq \emptyset$), so the space is free of pre-state failures.

We work on the quotient space induced by the constitutional invariants Φ_0:

$$\hat{S} = S/\equiv_{\Phi_0}$$

where $\equiv_{\Phi_0}$ identifies states with identical Φ_0-values. The reachability order descends to a partial order on $\hat{S}$. We denote the quotient map by $\pi : S \to \hat{S}$. All topological notions are defined on $\hat{S}$. Higher invariant levels Φ_i ($i > 0$) refine this quotient but do not change the underlying topology.

11.1 Poset from Reachability

Let $\leq$ be the reflexive-transitive closure of R_A on $\hat{S}$. Because cycles without new admissible information are inadmissible and Axiom 2 ensures such cycles collapse to the same Φ_0-class, $\leq$ is antisymmetric on $\hat{S}$. Hence $(\hat{S}, \leq)$ is a partially ordered set (poset).

11.2 Forward-Stable Sets and Topology

A set $U \subseteq \hat{S}$ is forward-stable under R_A iff it is upward-closed under $\leq$. The family $\mathcal{T}$ of all upward-closed subsets of $(\hat{S}, \leq)$ is an Alexandrov topology. We call $\mathcal{T}$ the admissible topology on $\hat{S}$.

11.3 Kernel Openness

For any non-empty successor set $S' \subseteq \hat{S}$, the base kernel can be expressed topologically as the intersection of Φ_0-classes:

$$K^{(0)}(S') = \bigcap_{\sigma' \in S'} [\sigma']_{\Phi_0}.$$

Operationally, v5.1.1 retains the consistency-based kernel definition of Section 2, which for Φ_0 coincides with this topological representation. Since each $[\sigma']_{\Phi_0}$ is upward-closed, and arbitrary intersections of upward-closed sets are upward-closed, $K^{(0)}(S')$ is open in $\mathcal{T}$.

11.4 Mode Refinement via Topology

The mode conditions of Section 5 can be understood topologically:

- Strict: $|S'| = 1$ implies the kernel is a single Φ_0-class, which is open.

- SAFE: all successors lie in the same Φ_0-class, so the kernel is that non-empty open class.
- DEFEASIBLE: successors belong to distinct Φ_0-classes; their intersection is empty, hence the kernel is $\emptyset$.

This topological perspective shows that the empty kernel in defeasible cases is not an error but a structural necessity: the system correctly refuses to posit common knowledge when admissible branches diverge in invariant terms.

12 Dynamic Updates as Topological Refinement

12.1 Update as Restriction

An update u selects a subset $\hat{S}_u \subseteq \hat{S}$ of states that survive. The new reachability relation is the restriction:

$$R_A^u = R_A \cap (\hat{S}_u \times \hat{S}_u).$$

The new poset order $\leq_u$ is the reflexive-transitive closure of R_A^u. If $\hat{S}_u = \emptyset$, the system halts with REFUSAL, preserving fail-closed.

12.2 Topological Interpretation

The admissible topology $\mathcal{T}$ consists of all upward-closed subsets of $(\hat{S}, \leq)$. For any open set $U \in \mathcal{T}$, U can be interpreted as the set of states compatible with current knowledge. The update restricts the state space to $\hat{S}_u$; the new topology $\mathcal{T}_u$ is the Alexandrov topology induced by $\leq_u$. The restriction operator

$$\mathrm{int}_u : \mathcal{T} \to \mathcal{T}_u, \quad \mathrm{int}_u(U) = U \cap \hat{S}_u$$

is decreasing, monotone, and idempotent.

12.3 Effect on Equivalences

After an update, both the set of states and the reachability structure are restricted. Consequently, the equivalence relations are recomputed on the new poset:

$$\sigma \sim_{\Phi_0}^u \tau \iff [\sigma]_{\Phi_0}^u = [\tau]_{\Phi_0}^u, \qquad \sigma \sim_Y^u \tau \iff \sigma \leq_u \tau \text{ and } \tau \leq_u \sigma.$$

Because $\hat{S}_u \subseteq \hat{S}$ and $\leq_u \subseteq \leq$, equivalence classes can only become smaller or disappear entirely. Thus updates make knowledge weakly sharper: they never coarsen distinctions.

12.4 Kernel and Modes After an Update

The kernel is recomputed on the new successor set. For a transition step with immediate successors $S' \subseteq \hat{S}_u$, we have

$$K^{(i),u}(S') = K^{(i)}(S') \cap \hat{S}_u,$$

where the intersection is taken after first restricting S' to $\hat{S}_u$ (i.e., S' is already understood as a subset of $\hat{S}_u$). Because the Φ_i-classes are open in $\mathcal{T}_u$, the kernel remains open in the refined topology. The mode classification is applied exactly as before, now relative to $\leq_u$ and $\sim^u_{\Phi_i}$.

12.5 Relation to Dynamic Epistemic Logic

In DEL, a public announcement ψ eliminates all worlds where ψ is false, which corresponds directly to a restriction $\hat{S}_u = \{\sigma \in \hat{S} \mid \sigma \models \psi\}$. The structural core of an update is therefore identical. While DEL is formulated over Kripke models, our framework operates on a poset induced by admissible reachability, suggesting a natural translation for acyclic, antisymmetric models.

13 Formal Properties: Lemmas and Theorems

We now consolidate the framework into a formal theory by stating and proving the key properties established in the preceding sections.

13.1 Order and Topology

Lemma 13.1 (Poset Structure)**.** *The reachability order $\leq$ defined as the reflexive-transitive closure of R_A on the quotient $\hat{S}$ is a partial order: reflexive, transitive, and antisymmetric.*

Proof. Reflexivity and transitivity hold by construction. Antisymmetry follows from Axiom 2: if $\sigma \leq \tau$ and $\tau \leq \sigma$ then $[\sigma]_{\Phi_0} = [\tau]_{\Phi_0}$. In the quotient this forces $\sigma = \tau$. □

Lemma 13.2 (Forward-stable = Upward-closed)**.** *A set $U \subseteq \hat{S}$ is forward-stable under R_A iff it is upward-closed under $\leq$.*

Proof. Forward-stability gives closure under single R_A-steps; by induction it extends to all finite paths, hence to $\leq$. Conversely, upward-closed under $\leq$ immediately implies closure under R_A because $R_A \subseteq \leq$. □

Theorem 13.3 (Openness of Φ_i-Classes in S)**.** *For any $\sigma \in S$, the set $[\sigma]_{\Phi_i} \subseteq S$ is open in the Alexandrov topology induced by $\leq$ on S.*

Proof. Take any $\sigma' \in [\sigma]_{\Phi_i}$ and any τ with $\sigma' \leq \tau$. By Axiom 1 and the definition of $\leq$, $\phi(\tau) = \phi(\sigma')$ for all $\phi \in \Phi_i$, so $\tau \in [\sigma]_{\Phi_i}$. Hence $[\sigma]_{\Phi_i}$ is upward-closed, and therefore open. □

Theorem 13.4 (Kernel Openness)**.** *For any non-empty successor set $S' \subseteq \hat{S}$, the kernel $K^{(0)}(S') = \bigcap_{\sigma' \in S'} [\sigma']_{\Phi_0}$ is open in $\mathcal{T}$.*

Proof. Each $[\sigma']_{\Phi_0}$ is upward-closed, and arbitrary intersections of upward-closed sets are upward-closed. Hence $K^{(0)}(S')$ is open. □

13.2 Admissibility Pipeline

Theorem 13.5 (RGR Failure)**.** *If a proposal q has empty relational component ($R = \emptyset$), then $A_{pre}(q) =$* Halt_Spec_Required*.*

Proof. The mapping $M : Q \to S \cup \{\perp\}$ preserves the relational component, so if $R = \emptyset$ the resulting tuple would have $R_\sigma = \emptyset$. By the completeness axiom, $\neg$Complete(σ). Hence the state cannot be formed, and the pre-state check returns Halt_Spec_Required. □

Theorem 13.6 (RCR Failure)**.** *If a state σ is relationally grounded ($R_\sigma \neq \emptyset$) but fails the invariant condition ($\Delta\Phi_0(\sigma) \neq 0$), then $A(q, \sigma) =$* Reject*.*

Proof. Directly from the definition of the Relational Closure Requirement. □

Theorem 13.7 (Pipeline Exhaustiveness)**.** *For any admissible proposal q and any state σ resulting from the mapping $M(q)$, exactly one of the following holds:*

- $A_{pre}(q) =$ Halt_Spec_Required *(RGR failure),*
- $A(q, \sigma) =$ Reject *(RCR failure),*
- $A(q, \sigma) =$ Admissible *(both gates satisfied).*

Thus the admissibility pipeline is exhaustive over the relevant cases.

Proof. The three cases are mutually exclusive by construction and cover all possibilities because either the proposal cannot be mapped to a complete state (RGR failure) or it can; in the latter case, either the invariants hold (RCR success) or they do not (RCR failure). □

13.3 Kernel and Mode Behaviour

Theorem 13.8 (Kernel Monotonicity)**.** *For all $i < j$:*

$$K^{(j)}(S') \subseteq K^{(i)}(S').$$

Proof. Because $\Phi_i \subseteq \Phi_j$, Φ_j-consistency implies Φ_i-consistency. Two cases:

- If S' is Φ_j-consistent, then it is also Φ_i-consistent, so $K^{(j)}(S') = S' = K^{(i)}(S')$.
- If S' is not Φ_j-consistent, then $K^{(j)}(S') = \emptyset \subseteq K^{(i)}(S')$.

Thus the inclusion holds in all cases. □

Theorem 13.9 (Strict Kernel)**.** *In Strict mode, $|S'| = 1$ and S' is Φ_i-consistent for all i. Hence $K^{(i)}(S') = S'$ for all i.*

Proof. With a unique model, all Φ_i-values are trivially constant across S'. Thus S' is Φ_i-consistent for every i. □

Theorem 13.10 (Safe Kernel)**.** *In Safe mode, $|S'| \geq 2$ and S' is Φ_i-consistent. Hence $K^{(i)}(S') = S'$.*

Proof. Immediate from the definition of Safe and the kernel definition. □

Theorem 13.11 (Defeasible Kernel)**.** *In Defeasible mode, S' is not Φ_i-consistent, so $K^{(i)}(S') = \emptyset$.*

Proof. Immediate from the definition of Defeasible and the kernel definition. □

13.4 Updates and Convergence

Lemma 13.12 (Restricted Φ_i-classes)**.** *For an update u with surviving state space $\hat{S}_u \subseteq \hat{S}$ and any $\sigma \in \hat{S}_u$:*

$$[\sigma]^u_{\Phi_i} = [\sigma]_{\Phi_i} \cap \hat{S}_u.$$

Proof. By definition, $[\sigma]^u_{\Phi_i}$ consists of those states $\tau \in \hat{S}_u$ that share all Φ_i-values with σ. This is exactly the intersection of the original class $[\sigma]_{\Phi_i}$ with $\hat{S}_u$. □

Theorem 13.13 (Kernel under Update)**.** *For an update u and any successor set $S' \subseteq \hat{S}_u$:*

$$K^{(i),u}(S') = K^{(i)}(S') \cap \hat{S}_u.$$

Proof.

$$K^{(i),u}(S') = \bigcap_{\sigma' \in S'} [\sigma']^u_{\Phi_i} = \bigcap_{\sigma' \in S'} ([\sigma']_{\Phi_i} \cap \hat{S}_u) = \left(\bigcap_{\sigma' \in S'} [\sigma']_{\Phi_i} \right) \cap \hat{S}_u = K^{(i)}(S') \cap \hat{S}_u.$$

□

Theorem 13.14 (Fail-closed Updates)**.** *If an update u leads to $\hat{S}_u = \emptyset$, the system halts with* Refusal*, preserving the fail-closed principle.*

Proof. By definition, a transition that would result in an empty state space is not admissible; the system returns Refusal. □

Theorem 13.15 (Continuity of Abstraction)**.** *Let $f : \hat{S} \to \hat{S}'$ be an order-preserving map between posets (i.e., $\sigma \leq \tau \Rightarrow f(\sigma) \leq' f(\tau)$) that respects the invariants: $\phi_i(\sigma) = \phi'_i(f(\sigma))$ for a corresponding invariant family Φ'. Then f is continuous with respect to the Alexandrov topologies.*

Proof. Take any open $U' \subseteq \hat{S}'$ (upward-closed in $\leq'$). We must show $f^{-1}(U')$ is upward-closed in $\leq$. Let $\sigma \in f^{-1}(U')$ and $\sigma \leq \tau$. Then $f(\sigma) \leq' f(\tau)$ by order-preservation, and $f(\sigma) \in U'$. Since U' is upward-closed, $f(\tau) \in U'$, so $\tau \in f^{-1}(U')$. Hence $f^{-1}(U')$ is upward-closed, i.e., open. □

Theorem 13.16 (Convergence under Iterative Updates)**.** *Let $u_1, u_2, \ldots$ be a sequence of updates such that $\hat{S}_{u_1} \supseteq \hat{S}_{u_2} \supseteq \cdots$ is a descending chain of subsets. If the chain stabilises after finitely many steps (i.e., the poset satisfies the descending chain condition on the subspace topology), then the kernels converge:*

$$\lim_{n \to \infty} K^{(i),u_n}(S'_n) = K^{(i),\lim}(S'_{\lim}),$$

where the limit is taken in the sense of eventual equality. In particular, if the chain is infinite and the descending chain condition holds, the chain must stabilise at a finite stage.

Proof. In a poset with the descending chain condition (DCC) on the underlying set, any descending chain of subsets eventually becomes constant. The kernels are subsets of these spaces, so they also stabilise. The limit kernel is then the kernel computed on the stable subspace. □

14 Categorical Extension of the Structural Layer

This section is largely unchanged from v4.9.1; it provides a categorical perspective on reachability and motivates enriched morphisms. The layered symbolic core does not alter this analysis, which remains an orthogonal addition.

14.1 Thin Categorical Reading of the Reachability Structure

The poset $(\hat{S}, \leq)$ defines a thin category $\mathcal{C}$:

- Objects: states $\sigma \in \hat{S}$
- Morphisms: there is a unique arrow $\tau \to \sigma$ iff $\tau \leq \sigma$

The category is thin because between any two objects there is at most one morphism.

The Yoneda embedding $y : \mathcal{C} \to [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ sends each state σ to the representable presheaf $y_\sigma = \mathrm{Hom}_{\mathcal{C}}(-, \sigma)$.

14.2 Collapse of Yoneda Equivalence in the Thin Case

Proposition 14.1 (Yoneda Equivalence Collapses)**.** *In the thin category $\mathcal{C}$ defined by the poset $(\hat{S}, \leq)$, two states $\sigma, \tau \in \hat{S}$ satisfy $y_\sigma \cong y_\tau$ if and only if $\sigma = \tau$.*

Proof. The Yoneda lemma gives $y_\sigma \cong y_\tau$ iff $\mathrm{Hom}_{\mathcal{C}}(-, \sigma) \cong \mathrm{Hom}_{\mathcal{C}}(-, \tau)$. In a thin category this isomorphism holds iff the two representable presheaves are equal as sets of objects, which happens precisely when the downward-closed sets of objects that can reach σ and τ coincide. Because the order is anti-symmetric, this forces $\sigma = \tau$. □

Consequently, structural equivalence $\sim_Y$ as defined via natural isomorphism of representables reduces to identity. Observational equivalence $\sim_{\Phi_0}$ is coarser; it can identify distinct states that share the same invariant values, even if they are incomparable in the order. This gap is what we earlier called a structural gap.

14.3 Why Collapse Matters: The Need for Richer Morphisms

The collapse theorem shows that in the current setting, where morphisms only encode reachability, the Yoneda embedding cannot distinguish states that are not identical. Therefore any non-trivial notion of semantic pair that goes beyond mere identity must be based on additional structure on morphisms, not on objects alone.

14.4 Candidate Enriched Framework

We propose to enrich the thin category $\mathcal{C}$ by replacing the set of morphisms (which is at most a singleton) with a set that carries additional data. A natural candidate is to let each admissible transition (σ, τ) (i.e., each $\sigma \leq \tau$) be decorated with:

- **Provenance**: a trace of the proposals and guards that generated the transition.

- **Admissibility profile**: which invariants were checked and how the mode (Strict, Safe, Defeasible) was determined.
- **Invariant signature**: the vector of Φ_i-values of the states involved.

Formally, we define an enriched category $\mathcal{C}_{\text{enr}}$ with the same objects as $\hat{S}$, but where the hom-set $\text{Hom}_{\text{enr}}(\sigma, \tau)$ is the set of all admissible paths from σ to τ together with their decorations. Composition is concatenation of paths (with merged decorations). The original thin category is recovered by forgetting the decorations.

14.5 Semantic Pairs in the Enriched Setting

In the enriched framework, we can define a semantic pair (σ, τ) as a pair of states that are connected by an invariant-enriched equivalence. One way to formalise this is to require the existence of enriched morphisms $f : \sigma \to \tau$ and $g : \tau \to \sigma$ such that $f \circ g$ and $g \circ f$ are equivalent to identity morphisms under some equivalence relation that respects the decorations.

Because the decorations carry detailed information about the transitions, two distinct states may become isomorphic if there exist admissible paths back and forth that are compatible with the invariant structure. For example, in a Safe mode scenario, all successors lie in the same Φ_i-class; if they are mutually reachable via admissible paths, they could be considered semantically equivalent under the enriched interpretation, even though they are distinct in the underlying poset.

14.6 Boundaries and Future Work

The enriched framework outlined here is deliberately sketched, not fully formalised. Its purpose is to demonstrate that the collapse of Yoneda equivalence in the thin case is not a weakness but a guide: it tells us where additional structure is needed. The following aspects remain for future work:

- Exact definition of the decoration algebra and composition laws.
- Compatibility with the invariants Φ_i and the kernel $K^{(i)}$.
- Proof that enriched equivalence indeed yields a non-trivial relation that refines observational equivalence.
- Application to multi-agent scenarios where different agents may have different admissible path decorations.

Crucially, the kernel $K^{(i)}$ remains the set-theoretic consistency-based object; it is not replaced by a colimit or any categorical construction. The enriched layer is an orthogonal addition that does not alter the core fail-closed validation pipeline.

15 Execution Semantics

The framework is mechanised using a satisfiability modulo theories (SMT) solver, e.g., Z3. The following steps define the execution flow:

1. **Input**
   - A symbolic state space Ψ over a set of variables V.
   - A hierarchy $\Phi_0 \subseteq \cdots \subseteq \Phi_k$ where each Φ_i is a set of formulas over V.
   - A successor constraint Ξ (from a proposal q).
2. **Constitutional checks**
   - Verify that Ψ and Ξ are well-formed and use only the declared logic.
   - Check that $\Psi(\sigma) \Rightarrow R_\sigma \neq \emptyset$ (RGR).
   - If any check fails $\rightarrow$ Halt_Spec_Required.
3. **Consistency check per level** For each level i from k down to 0:
   - Determine whether the substate $S' = \{\sigma \mid \Psi(\sigma) \wedge \Xi(\sigma)\}$ is Φ_i-consistent by testing if there exist two models of $\Psi \wedge \Xi$ that differ on some $\phi \in \Phi_i$.
   - If no such pair exists, the substate is Φ_i-consistent $\rightarrow$ kernel $K^{(i)}(S') = S'$.
4. **Admissibility** Let $I^* = \max\{i \mid S' \text{ is } \Phi_i\text{-consistent}\}$.
   - If I^* exists $\rightarrow$ Admissible_Level(I^*); return I^* and the kernel formula $\Psi \wedge \Xi$.
   - Otherwise $\rightarrow$ Refusal.
5. **Mode profile** For each level i:
   - Strict: if $\Psi \wedge \Xi$ has exactly one model (uniqueness check via solver).
   - Safe: if S' is Φ_i-consistent and has at least two models.
   - Defeasible: otherwise.

   The profile is descriptive only; admissibility is determined solely by I^*.
6. **Solver interaction**
   - All logical checks are implemented via SMT solver calls.

- If a query returns UNKNOWN (e.g., due to undecidability or timeout) → REFUSAL.
- All formulas are restricted to a decidable logic (e.g., QF_LIA, QF_UF).

This execution semantics guarantees that the framework is **mechanically executable** and **deterministic**, adhering to the fail-closed principle.

16 Conclusion

16.1 Core Insights

Epistemic logic is replaced by constraint-based state evolution. Knowledge is not belief, but invariant set under admissible transitions. Uncertainty is not probability, but emptiness at higher invariant levels. Reasoning is restriction, not generation.

16.2 System Properties

- Fail-closed
- No implicit assumptions
- Existence equivalent to admissibility
- Knowledge formation tied to level-relative kernels $K^{(i)}$
- Controlled branching through Strict, Safe, and Defeasible modes per level
- Knowledge gradation via invariant hierarchy
- Multi-agent constraints for communication, consistency, and privacy
- Paradox prevention within admissible state space
- Topological grounding: the admissible state space carries an Alexandrov topology in which kernel sets are open
- Relational admissibility: the two sequential gates RGR and RCR define a complete pipeline
- Dynamic updates: knowledge sharpens (never coarsens) via restriction of state space and reachability
- Formal theorem collection: all core properties are proved from the axioms

- Categorical extension: a thin category of reachability collapses Yoneda equivalence to identity, motivating enriched morphisms for non-trivial semantic pairs

16.3 Final Statement

The system does not solve classical paradoxes. It prevents their construction within the admissible state space.

16.4 Core Principles

Truth is what survives all admissible transformations.
Reasoning is restriction, not generation.
Ambiguity is structural failure.
A system is reliable by what it refuses.
No interpretation. No implicit meaning. No assumptions.
Only: definition → rule → consequence

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All core concepts, definitions, and architectural principles originate from the author. The role of these systems was strictly assistive: to test consistency, expose implicit assumptions, and accelerate the formalization process.

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A Python/Z3 Reference Implementation

The reference implementation is available in the repository at `src/gsa51_core.py`. The following key functions implement the constitutional core:

- `_check_consistency(xi, level)` – checks if S' is Φ_i-consistent.
- `kernel(xi, level)` – returns $(nonempty, formula)$.

- `admissibility_level(xi)` – returns $(I^*, kernel_formula)$.
- `invariant_signature(model, level)` – returns the Φ_i-values.
- `evaluate(xi)` – high-level fail-closed evaluation.
- `is_unique(xi)` – checks for exactly one model (Strict mode).

The implementation follows the execution semantics described in Section 15. All solver calls are deterministic, and unknown results are treated as Refusal.